

THE SARKISOV PROGRAM

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ABSTRACT. Let $\phi: (X, B_X) \dashrightarrow S$ and $\phi': (X', B_{X'}) \dashrightarrow S''$ be two Mori fiber spaces which are the output of a minimal model program for a kawamata log terminal pair (W, B_W) . Then ϕ' may be obtained from ϕ by a finite sequence of Sarkisov links.

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1. INTRODUCTION

Recall the following:

Conjecture 1.1. *Let (Z, Ξ) be a kawamata log terminal pair.*

Then we may run a $(K_Z + \Xi)$ -MMP so $p: Z \dashrightarrow X$ such that either

- (1) *(X, Δ) is a minimal model, that is $K_X + \Delta$ is nef, or*
- (2) *there is a Mori fibre space $\phi: X \dashrightarrow S$, that is $\rho(X/S) = 1$ and $-(K_X + \Delta)$ is ϕ -ample,*

where $\Delta = p_\Xi$.*

By [1] the only case of (1.1) which is not known is when $K_Z + \Xi$ is pseudo-effective but neither Ξ nor $K_Z + \Xi$ is big. Unfortunately the output is unique in neither case (1) or case (2) of (1.1). It is proved in [1] that the number of minimal models is finite if $K_X + \Delta$ is big, and it is conjectured that in general the number of minimal models is finite up to automorphisms.

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Moreover, recently Kawamata, see [3], has proved:

Theorem 1.2. *Suppose that (X, Δ) and (Y, Γ) are two minimal models obtained from the same pair (Z, Ξ) .*

Then the birational map $g: X \dashrightarrow Y$ may be factored as a sequence of $(K_Z + \Xi)$ -flops.

The Sarkisov program establishes a similar result to (1.2) in the case of Mori fibre spaces:

Theorem 1.3. *Let (Z, Φ) be a projective $\mathbb{Q}$ -factorial kawamata log terminal pair. Let $\phi: X \rightarrow U$ and $\psi: Y \rightarrow V$ be two Mori fibre spaces, which are both outcomes of running the $(K_Z + \Phi)$ -MMP.*

Then $g: X \dashrightarrow Y$ is the composition of Sarkisov links.

Note that if X and Y are birational and the pairs (X, Δ) and (Y, Γ) both have $\mathbb{Q}$ -factorial terminal singularities, then ϕ and ψ are automatically the outcome of running the $(K_Z + \Phi)$ -MMP:

Corollary 1.4. *Suppose that $(X, \Delta) \rightarrow S$ and $(Y, \Gamma) \rightarrow T$ are both Mori fibre spaces, where (X, Δ) and (Y, Γ) are two $\mathbb{Q}$ -factorial terminal pairs.*

Then X and Y are birational iff they are related by a sequence of Sarkisov links.

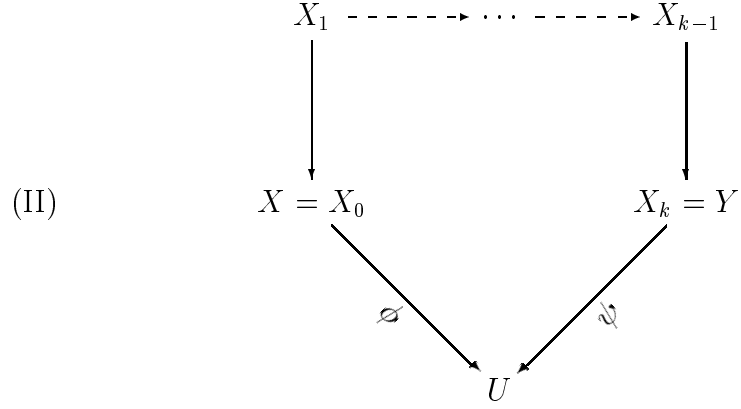
In fact the most interesting case of (1.3) is when $\Delta = \Gamma = 0$ and both X and Y have terminal singularities.

We recall the definition of a Sarkisov link. Suppose that $\phi: X \rightarrow U$ and $\psi: Y \rightarrow V$ are two Mori fibre spaces. A Sarkisov link between ϕ and ψ is one of four types:

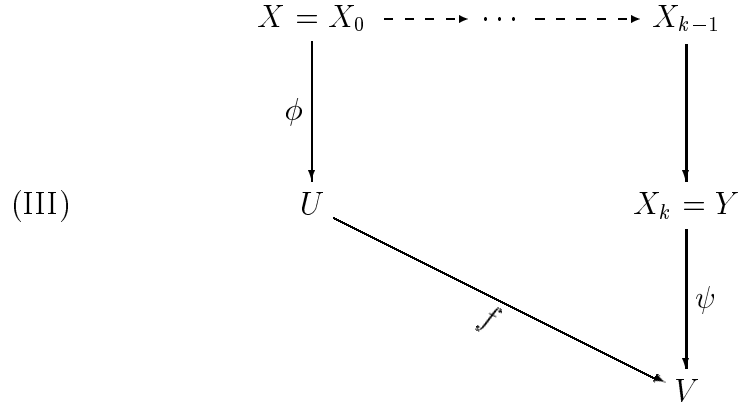
$$(I) \quad \begin{array}{ccc} X_1 & \dashrightarrow \cdots \dashrightarrow & X_k = Y \\ \downarrow & & \downarrow \psi \\ X = X_0 & & V \\ \downarrow \phi & \nearrow g & \\ U & & \end{array}$$

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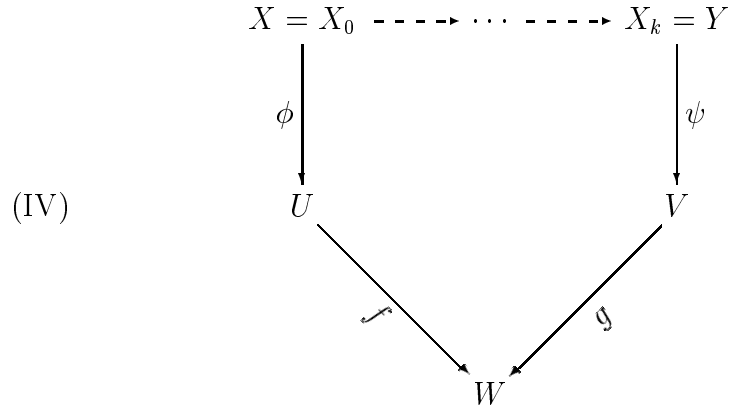
where $X_1 \longrightarrow X$ is an extremal divisorial contraction, $X_1 \dashrightarrow X_k = Y$ is a sequence of flops and $\rho(V/U) = 1$.



where both $X_1 \longrightarrow X_0 = X$ and $X_{k-1} \longrightarrow X_k = Y$ are extremal divisorial contractions and $X_1 \dashrightarrow X_{k-1}$ is a sequence of flops.



where $X = X_0 \dashrightarrow X_{k-1}$ is a sequence of flops, $X_{k-1} \longrightarrow X_k = Y$ is an extremal divisorial contraction and $\rho(U/V) = 1$.



where $X = X_0 \dashrightarrow X_k = Y$ is a sequence of flops and $\rho(U/W) = \rho(V/W) = 1$.

To explain the origins of this program, let us start with the well-known case of surfaces. If $\phi: X \longrightarrow U$ is a Mori fibre space and X is a smooth projective surface, then either U is a point and $X = \mathbb{P}^2$ or ϕ is a $\mathbb{P}^1$ -bundle. Now suppose that $\psi: Y \longrightarrow V$ is another Mori fibre space and let $g: X \dashrightarrow Y$ be a birational map. If U and V are both curves of genus at least one, then it is well-known that g is the composition of a sequence of elementary transformations. Recall that an elementary transformation is obtained by picking a point, blowing it up and then contracting the old fibre. Note that all of these transform

If X is rational, then one must allow two other types of links, namely blowing up a point in $\mathbb{P}^2$, which replaces the Mori fibre

The proof is based on the original ideas of the Sarkisov program (as explained by Corti and Reid [2]). The main twist is that we are unable to show termination of an arbitrary sequence of flips, nor the acc property for pseudo-effective thresholds. Instead our argument is based on the principle of finiteness of minimal models for kawamata log terminal pairs (Z, Θ) such that Θ varies in a compact subset of the big cone.

2. NOTATION AND CONVENTIONS

We work over the field of complex numbers $\mathbb{C}$. A $\mathbb{R}$ -Cartier divisor D on a normal variety X is *nef* if $D \cdot C \geq 0$ for any curve $C \subset X$. We say that two $\mathbb{R}$ -divisors D_1, D_2 are $\mathbb{R}$ -linearly equivalent ($D_1 \sim_{\mathbb{R}} D_2$) if $D_1 - D_2 = \sum r_i(f_i)$ where $r_i \in \mathbb{R}$ and f_i are rational functions on X . We say that a $\mathbb{R}$ -Weil divisor D is *big* if we may find an ample $\mathbb{R}$ -divisor A and an effective $\mathbb{R}$ -divisor B , such that $D \sim_{\mathbb{R}} A + B$. A divisor D is *pseudo-effective*, if for any ample divisor A and any rational number $\epsilon > 0$, the divisor $D + \epsilon A$ is big. If A is a $\mathbb{Q}$ -divisor, we say that A is a *general ample divisor* if A is ample and there is a sufficiently divisible integer $M > 0$ such that MA is very ample and $MA \in |MA|$ is very general. If A is a $\mathbb{R}$ -divisor, we say that A is a *general ample divisor* if $A = \sum r_i A_i$ where $r_i \in \mathbb{R}_{\geq 0}$ and A_i are general ample $\mathbb{Q}$ -divisors.

A *log pair* (X, Δ) is a normal variety X and an effective $\mathbb{R}$ -Weil divisor Δ such that $K_X + \Delta$ is $\mathbb{R}$ -Cartier. We say that a log pair (X, Δ) is *log smooth*, if X is smooth and the support of Δ is a divisor with global normal crossings. A projective morphism $g: Y \longrightarrow X$ is a *log resolution* of the pair (X, Δ) if Y is smooth and $g^{-1}(\Delta) \cup \{\text{exceptional set of } g\}$ is a divisor with normal crossings support. We write $g^*(K_X + \Delta) = K_Y + \Gamma$ and $\Gamma = \sum b_i \Gamma_i$ where Γ_i are distinct reduced irreducible divisors and $g_* \Gamma = \Delta$. The discrepancy of Γ_i is $a(\Gamma_i, X, \Delta) = -b_i$.

A pair (X, Δ) is *kawamata log terminal* (KLT) if $b_i < 1$ for all i . We say that the pair (X, Δ) is *log canonical* if $b_i \leq 1$ for all i . We say that the pair (X, Δ) is *terminal* if the discrepancy of any exceptional divisor is greater than zero.

3. AMPLE MODELS

We recall some definitions and notation from [1]:

Definition 3.1. Let Z be a $\mathbb{Q}$ -factorial projective variety and suppose that $K_Z + \Theta$ is kawamata log terminal. Suppose that $f: Z \dashrightarrow U$ is a rational map. We say that f is an **ample model** for (Z, Θ) if there is a log terminal model $p: Z \dashrightarrow X$ for $K_X + \Theta$, a morphism $\phi: X \rightarrow U$ and an ample divisor C on U , such that $K_X + \Delta = \phi^*C$, where $\Delta = \phi_*\Theta$.

Definition 3.2. Let Z be a normal variety and let V_0 be a finite dimensional affine subspace of the space of all Weil divisors. Define

$$\mathcal{L}(V_0) = \{ \Theta \in V_0 \mid K_Z + \Theta \text{ is log canonical} \}.$$

Moreover, fixing a $\mathbb{Q}$ -divisor $\Xi = A + B$, where A is ample and $B \geq 0$, whose support has no common components with any element of Λ , define

$$\begin{aligned} V_{0,\Xi} &= \{ \Theta \mid \Theta - \Xi \geq 0 \in V_0 \}, \\ \mathcal{L}_\Xi(V_0) &= \{ \Theta \in V_{0,\Xi} \mid K_Z + \Theta \text{ is log canonical} \}, \\ \mathcal{P}_\Xi(V_0) &= \{ \Theta \in \mathcal{L}_\Xi \mid K_Z + \Theta \text{ is pseudo-effective over } U \}. \end{aligned}$$

Given a rational map $f: Z \dashrightarrow U$, define

$$\mathcal{P}_f = \{ \Theta \in \mathcal{P}_\Xi(V_0) \mid f \text{ is the ample model of } (Z, \Theta) \}.$$

Theorem 3.3. Let $(Z, \Xi = A + B)$ be a projective variety with $\mathbb{Q}$ -factorial kawamata log terminal singularities, where A is ample and $B \geq 0$. Let V_0 be a finite dimensional affine subspace of the space of all Weil divisors.

Then there are finitely many rational maps $f_i: Z \dashrightarrow U_i$, $1 \leq i \leq m$, with the following properties:

- (1) the sets $\mathcal{P}_i = \mathcal{P}_{f_i}$ are a partition of $\mathcal{P}_\Xi(V_0)$ and each $\overline{\mathcal{P}}_i$ is a rational polytope.

- (2) If $\mathcal{P}_k \cap \overline{\mathcal{P}_j} \neq \emptyset$ then there is a contraction morphism $p_{j,k}: U_j \longrightarrow U_k$ which fits into a commutative diagram

$$\begin{array}{ccc} Z & \xrightarrow{f_j} & U_j \\ & \searrow \pi_{j,k} & \downarrow p_{j,k} \\ & & U_k. \end{array}$$

Proof. (1) is proved in [1].

Pick $\Theta \in \mathcal{P}_k \cap \overline{\mathcal{P}_j}$ and $\Theta' \in \mathcal{P}_j$. Then

$$\Theta_t = \Theta + t(\Theta' - \Theta) \in \mathcal{P}_j \quad \text{iff} \quad t \in (0, 1].$$

By definition of an ample model, there is a birational map $p_j: Z \dashrightarrow X_j$, whose inverse does not contract any divisors and a contraction morphism $\phi_j: X_j \longrightarrow U_j$ such that if we put $\Delta = p_{j*}\Theta$ and $\Delta' = p_{j*}\Theta'$ and

$$\Delta_t = p_{j*}\Theta_t = \Delta + t(\Delta' - \Delta),$$

then $K_{X_j} + \Delta_t$ is log terminal and there is an ample divisor H_t on U_j such that

$$K_{X_j} + \Delta_t = \phi_j^* H_t,$$

for all $t > 0$. It follows that

$$K_{X_j} + \Delta = \phi_j^* H_0 \quad \text{where} \quad H_0 = 2H_{1/2} - H_1,$$

is nef. Since we may find a big divisor Δ'' such that

$$K_{X_j} + \Delta \sim_{\mathbb{Q}} K_{X_j} + \Delta'',$$

is kawamata log terminal, $K_{X_j} + \Delta$ is semiample by the base point free theorem. Thus it defines a contraction morphism $X_j \longrightarrow U_k$. But then H_0 is also semiample and so it also defines a contraction morphism $\pi_{k,j}: U_j \longrightarrow U_k$ which factors $f_k = \pi_{k,j} \circ f_j$. $\square$

Theorem 3.4. *Notation as in (3.3). Suppose that the elements of V_0 span the Néron-Severi group of Z .*

- (1) *Pick an index j such that $\mathcal{P}_j$ has maximal dimension. Then $f_j: Z \dashrightarrow U_j$ is birational and U_j is $\mathbb{Q}$ -factorial.*
- (2) *Pick an index k such that $\mathcal{P}_k$ is a polytope of codimension one. Assume that $f_k: Z \dashrightarrow U_k$ is not birational and that $\mathcal{P}_k \subset \overline{\mathcal{P}_j}$. Then the induced morphism $\pi = \pi_{j,k}: X \longrightarrow U$ has relative Picard number one.*

- (3) *Pick an index l such that $\mathcal{P}_l$ is a polytope of codimension two. Assume that the ample model $t: Z \dashrightarrow W$ associated to $\mathcal{P}_l$ is not birational and that $\mathcal{P}_l \subset \overline{\mathcal{P}_j}$. Then morphism $\xi = \pi_{l,k}: X \rightarrow W$ has relative Picard number at most two with equality iff $\mathcal{P}_k$ is not a subset of $\overline{\mathcal{P}_j}$.*

Proof. For each $a \in \{j, k, l\}$, pick $\Theta_a \in \mathcal{P}_a$ belonging to the interior. As $\mathcal{P}_j$ is of maximal dimension and V_0 contains an ample divisor, we may pick $H \in V_0$ ample such that $\Theta_j + H \in \mathcal{P}_j$. As $K_Z + \Theta_j + H$ is big, and the ample model of $K_Z + \Theta_j$ is equal to the ample model of $K_Z + \Theta_j + H$, it follows that f_j is birational.

Let $u: X \rightarrow U_j$ be a log terminal model of U and let $f = u^{-1} \circ f_j: Z \dashrightarrow X$. Let $\Delta_j = f_{j*}\Theta$ and $\Delta = f_*\Theta$. As U has log terminal singularities it follows that u is small. But then as

$$K_X + \Delta = u^*(K_{X_j} + \Delta_j),$$

it follows that $K_X + \Delta$ is kawamata log terminal. Let H_1 be an ample divisor on X . Possibly replacing H_1 by a smaller multiple, we may find $H \in V_0$ such that $\Theta_j + H \in \mathcal{P}_j$ where H is numerically equivalent to f^*H_1 . But then X is the ample model of $K_Z + \Theta + H$, u is the identity and so U is $\mathbb{Q}$ -factorial. This is (1). $\square$

4. THE MAIN RESULT

Lemma 4.1. *Let $p: Z \dashrightarrow X$ be a birational map between projective $\mathbb{Q}$ -factorial varieties, whose inverse does not contract any divisors.*

If p is $(K_Z + \Xi)$ -negative and $K_Z + \Xi$ is kawamata log terminal then X is a possible outcome of the $(K_Z + \Xi)$ -MMP over X

Proof. Pick H ample on Z so that $K_Z + \Xi + H$ is ample and kawamata log terminal. Then the $(K_Z + \Xi)$ -MMP with scaling of H terminates with a log terminal model X' over X . As p is $(K_Z + \Xi)$ -negative then $X' \rightarrow X$ is small and as X is $\mathbb{Q}$ -factorial, $X' = X$. $\square$

Lemma 4.2. *Let $\phi: X \rightarrow U$ and $\psi: Y \rightarrow V$ be two Sarkisov related Mori fibre spaces for $K_X + \Delta$ and $K_Y + \Gamma$.*

Then we may find a log pair $(Z, \Xi = A + B)$ and two divisors $0 \leq G$ and $0 \leq H$ such that

- $(Z, \Xi + G + H)$ is log smooth,
- Z is projective,
- A is ample and $B \geq 0$,
- $K_Z + \Xi + G + H$ is kawamata log terminal,
- the components of $H + G$ span the Néron-Severi group of Z ,

- the induced birational maps $p: Z \dashrightarrow X$ and $q: Z \dashrightarrow Y$ are morphisms,
- $\phi \circ p$ and $\psi \circ q$ are ample models of $(Z, \Xi + G)$ and $(Z, \Xi + H)$,
- if $h + g \leq 1$ and $\phi_1: X_1 \rightarrow U_1$ is an ample model of $(Z, \Xi + gG + hH)$, then ϕ_1 is a possible outcome of the $(K_Z + \Xi)$ -MMP.

In particular ϕ and ψ are two possible outcomes of the $(K_Z + \Xi)$ -MMP.

Proof. By assumption we may find a $\mathbb{Q}$ -factorial kawamata log terminal pair (Z, Φ) and two birational morphisms $p: Z \dashrightarrow X$ and $q: Z \dashrightarrow Y$ such that ϕ and ψ are two possible outcomes of the $(K_Z + \Phi)$ -MMP. Let $f: S \rightarrow Z$ be any log resolution of (Z, Φ) which resolves the indeterminacy of p and q . We may write

$$K_S + \Psi = f^*(K_Z + \Phi) + E,$$

where E and Ψ are effective, with no common components, E is exceptional and $f_*\Psi = \Phi$. Pick any effective divisor F with support equal to the full exceptional locus such that $K_S + \Psi + F$ is kawamata log terminal. As f is $(K_S + \Phi + F)$ -negative, $K_Z + \Phi$ is kawamata log terminal and Z is $\mathbb{Q}$ -factorial, the $(K_S + \Psi + F)$ -MMP over Z terminates with the pair (Z, Φ) by (4.1).

Thus replacing (Z, Φ) with $(S, \Psi + F)$ we may assume that p and q are morphisms, and the support of Φ has normal crossings. Let A be any ample $\mathbb{Q}$ -divisor on Z . Pick a sufficiently ample $\mathbb{Q}$ -divisor C on U such that

$$-(K_X + \Delta) + \psi^*C,$$

is ample. Then we may find an ample and effective $\mathbb{Q}$ -divisor G_1 such that

$$K_X + \Delta + G_1 = \psi^*C,$$

is kawamata log terminal and p^*G_1 is equal to the strict transform of G_1 . As p^*G_1 is big and semiample, possibly replacing A by a small multiple, we may assume that there is a divisor G_2 with $2G_2 \sim p^*G_1$ such that $G_2 - A \geq 0$ and

$$K_Z + A + \Psi + G = K_Z + A + \Psi + G_2 + 1/2p^*G_1 \sim_{\mathbb{Q}} K_Z + \Psi + p^*G_1,$$

is kawamata log terminal. But then ϕ is the ample model of $(Z, \Xi + G)$. Similarly we may pick H so that ψ is the ample model of $(Z, \Xi + G)$. Since H_1 and $G_1 = q_*G$ are general, the support of $\Xi + H + G$ has normal crossings and $(Z, \Xi + H + G)$ is kawamata log terminal. $\square$

Definition 4.3. *Notation as in (4.2). Let V be the affine space spanned by $\Xi + H$ and $\Xi + G$.*

*We say a point Θ on the boundary of $\mathcal{P}_{\Xi}$ is **inner** if $K_Z + \Theta$ is not big.*

Let Θ be a point of the inner boundary. We associate the following data to Θ :

Enumerate $\mathcal{Q}_0, \mathcal{Q}_1, \dots, \mathcal{Q}_k$, in an anticlockwise direction, the polytopes of dimension two whose closure contains Θ . Let X_i be the ample model corresponding to any point of $\mathcal{Q}_i$. Then there are contraction morphisms $\pi_i = \pi_{X_i/W}: X_i \rightarrow W$. Let $X = X_0$, $Y = Y_k$, $\phi = \pi_0: X \rightarrow U$ and $\psi = \pi_k: Y \rightarrow V$.

Let $\mathcal{Q}$ and $\mathcal{Q}'$ be the two polytopes of dimension one whose closure contains Θ . We suppose that $\mathcal{Q}$ is in the closure of $\mathcal{Q}_0$ and $\mathcal{Q}'$ is in the closure of $\mathcal{Q}_k$. Let U be the ample model to any point of $\mathcal{Q}$ and V be the ample model corresponding to any point of $\mathcal{Q}'$. Then there are contraction morphisms $f = \pi_{U/W}: U \rightarrow W$ and $g = \pi_{V/W}: V \rightarrow W$. Finally note that there are birational maps $h_i: X_i \dashrightarrow X_{i+1}$ such that $\pi_i = \pi_{i+1} \circ h_i$. We say that a point Θ on the inner boundary is a **turning point** if $k > 0$.

Corollary 4.4. *Notation as in (4.2) and (4.3).*

Then we may pick H_1 and G_1 such that for all points Θ on the inner boundary

- (1) X_i is $\mathbb{Q}$ -factorial, $0 \leq i \leq k$ and
- (2) $\rho(X_i/W) \leq 2$ with equality iff $0 < i < k$.

Proof. Just pick H and G so that we get a general slice of $\mathcal{P}_\Xi$ and apply (3.4). $\square$

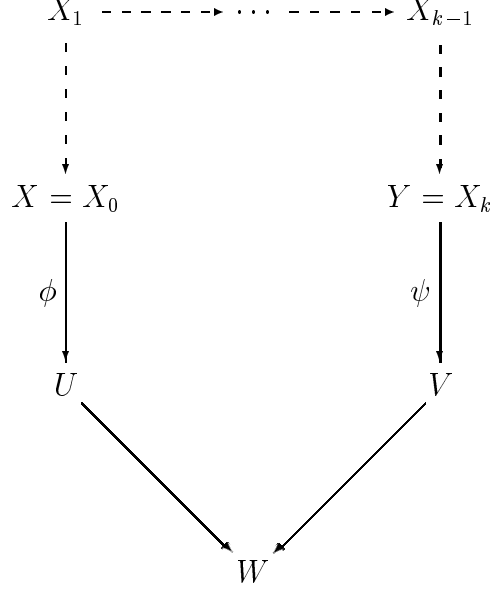
Theorem 4.5. *Notation as in (4.2) and (4.3). Assume that we have generic behaviour. Let $\Theta \in \mathcal{P}_\Xi$ be a turning point and let e be the cardinality of the set $\{U, V, W\}$.*

Then we have

- (1) $e = 1$ iff we have a link of type II,
- (2) $e = 2$ iff we have a link of type I or III,
- (3) $e = 3$ iff we have a link of type IV.

Proof. In each case, one direction is clear; the type of the link determines e .

Note that we have a diagram:



Note that $h_i: X_i \dashrightarrow X_{i+1}$ is either an extremal divisorial contraction, or an extraction divisorial extraction or a flop. Since X_i and X_{i+1} have the same Picard number for $0 < i < k$, $h_i: X_i \dashrightarrow X_{i+1}$ is a flop for $0 < i < k$.

Since $\phi: X \rightarrow U$ is extremal, it follows that either the birational map $h_1: X_1 \dashrightarrow X_0$ is an extremal divisorial contraction or $f: U \rightarrow W$ is not an isomorphism and h_1 is a flop. We have a similar dichotomy for $h_k: Y = X_k \rightarrow X_{k-1}$ and $g: V \rightarrow W$.

Suppose that $e = 1$. Then f and g are isomorphisms and $h_1: X_1 \rightarrow X_0 = X$ and $h_k: X_{k-1} \rightarrow X_k = Y$ are divisorial extremal contractions. This is a link of type II.

Suppose that $e = 2$. Possibly switching X and Y , we may assume that $f: U \rightarrow W$ is an isomorphism. But then $h_1: X_1 \rightarrow X_0 = X$ is an extremal divisorial extraction and otherwise the birational maps $h_i: X_i \dashrightarrow X_{i+1}$ are flops. Thus we have a link of type I or III.

Now suppose that $e = 3$. Then neither f nor g is an isomorphism so that h_1 and h_k are both flops. This is a link of type IV. $\square$

Proof of (1.3). We pick a model whose existence is guaranteed by (4.2) and (3.4), and we use the notation of (4.3).

Imagine tracing the path given by the inner boundary. We start at the point $(1, 0)$ and we proceed towards the point $(0, 1)$. Every turning point corresponds to a Sarkisov link $g_i: X(i) \rightarrow Y(i) = X(i+1)$, by (4.5), and it is clear that g is then the composition of $g_1, g_2, \dots, g_{m-1}$ and g_m . $\square$

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